

Assignment 1

Assignment Topics:

Agents, Types of agents, Uninformed Searches and informed searches

Guidelines:

1. Deadlines should be kept in mind. No extension in assignment dates would be given. No late submission will be accepted.
2. This is an individual assignment. **PLAGIARISM IS NOT ACCEPTABLE AT ALL! Zero marks will be given in case of plagiarism.**
3. **Deadline = 02-07-2025, Deadline is hard and firm.**

Question No 01:

A semi-autonomous robot named ARES is deployed to explore a newly discovered region of Mars suspected to harbor microbial life. ARES must:

- Navigate uneven, cratered terrain with low gravity and frequent dust storms.
- Collect geological and chemical samples from high-interest zones.
- Avoid environmental hazards like deep craters, sharp rocks, and radiation.
- Operate independently for extended periods with intermittent communication to Earth.
- Make adaptive decisions based on energy reserves, limited memory, sensor input, and terrain.

Your Task:

1. Provide a complete PEAS (Performance measure, Environment, Actuators, Sensors) description of the ARES mission in tabular form.
2. Environment Classification (20 marks)

For the ARES environment, classify and justify each of the following six properties:

Property	Choose One	Justification (2–3 lines)
a. Observability	Fully / Partially	
b. Determinism	Deterministic / Stochastic	
c. Episodic Nature	Episodic / Sequential	
d. Dynamics	Static / Dynamic	
e. State Structure	Discrete / Continuous	
f. Agent Multiplicity	Single-agent / Multi-agent	

Question No 02:

Pick a real-world example of your choice (a robot or software agent/bot, except for the examples discussed in detail in the class e.g., taxi & vacuum-cleaner agents) and describe different scenarios in the context of the following agent types that solve the problem.

- i. Reflex agent
- ii. Model-based
- iii. Goal-based
- iv. Utility-based
- v. Learning agent

Question No 03:

A problem can be described as a “search problem” by specifying the following components: (a) initial state, (b) actions, (c) transition model/successor function, (d) goal state, (e) path cost (assume a path cost of 1 per action, if not specified explicitly). For each of the following problems, identify the above components.

- I. There are six glass boxes in a row, each with a lock. Each of the first five boxes holds a key unlocking the next box in line, the last holds a banana. You have the key to the first box, and you want the banana.
- II. Consider a 3-coins problem, C1-C2-C3 where each coin has two sides: a head (or H) and a tail (or T). Only one coin can be flipped at a time which will change an H to T or a T to H. Since, we have three coins, the coin position would tell the coin to be flipped, that is, C1 will flip coin 1 and so on. The initial state is H-H-H and the goal is to get at least two tails.
- III. In the “missionaries and cannibals” problem, three missionaries and three cannibals must cross a river using a boat which can carry at most two people, under the constraint that, for both banks, if there are missionaries present on the bank, they cannot be outnumbered by cannibals (if they were, the cannibals would eat the missionaries). The boat cannot cross the river by itself with no people on board.

Question No 04:

You are developing an AI-based rescue agent to navigate an alien environment represented as a weighted graph. The agent starts from the base station 'S' and must reach the target rover location 'G'. However, the terrain is treacherous:

- Some paths are misleading but cheap.
- Some regions are deeply nested.
- Some paths lead into cycles or dead ends.
- Visibility is limited, so only uninformed search can be used.

Part A: Design a Custom Large Graph

Design your own graph in code (as an adjacency list or dictionary of dictionaries). It must:

- Contain at least more than 12 nodes (e.g., 'S', 'A', 'B', ..., 'L', 'G').
- Include non-uniform edge costs (e.g., 1, 2, 7, 25, etc.)
- Include:
 - At least one misleading low-cost path that doesn't reach 'G'.
 - At least one cycle or deep subtree (for DLS difficulty).
 - Start node labeled 'S' and goal node labeled 'G'.

Part B: Implement the following uninformed search with both with and without expanded list:

1. Uniform-Cost Search (UCS)
2. Depth-Limited Search (DLS)
 - You must choose the limit thoughtfully.
 - Your implementation should detect if the goal was not reachable within the depth limit.
3. Iterative Deeping search
4. BFS
5. DFS

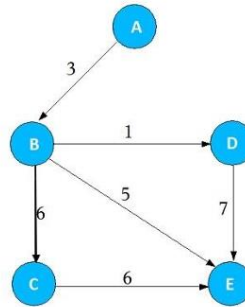
Each implementation must output:

- Path from 'S' to 'G'
- Total cost (UCS only)
- Nodes expanded

Question No 05:

Consider the graph with A as the start node and E as the goal node with heuristic values given in the table.

$h(A)$	8
$h(B)$	7
$h(C)$	7
$h(D)$	5
$h(E)$	0



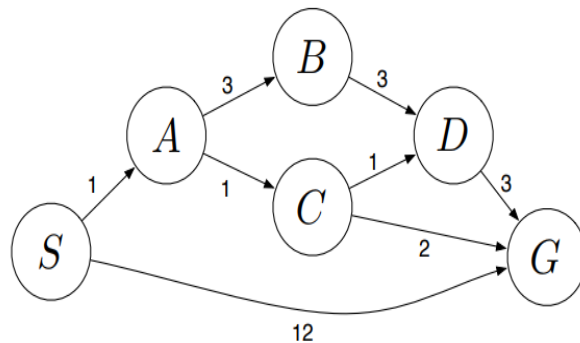
Answer the following question:

1. All heuristics are admissible. If not, make them admissible.
2. All heuristics are consistency. If not, make them consistency.
3. Run A* algorithm using the consistent heuristic values, break ties alphabetically. Clearly show the nodes on the open-list, closed-list, and final path with their f-value.

Question No 06:

Consider the following graph and heuristics given in table

State	h_1	h_2
S	5	4
A	3	2
B	6	6
C	2	1
D	3	3
G	0	0

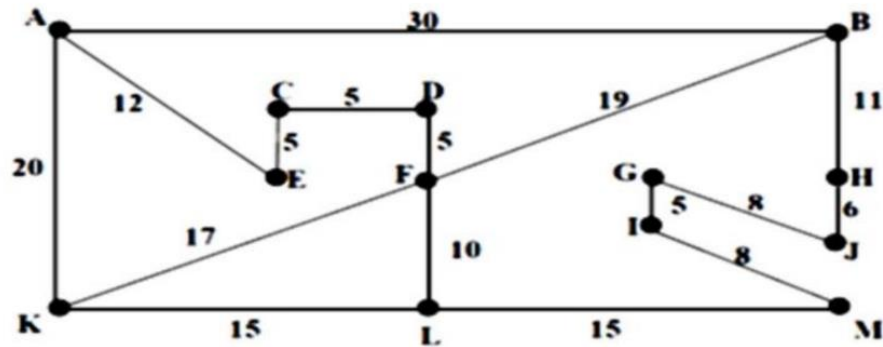


Answer the following questions, also provide the justification

- | | | |
|------|---------------------------|--------|
| i. | Is h_1 admissible? | Yes/No |
| ii. | Is h_1 consistent? | Yes/No |
| iii. | Is h_2 admissible? | Yes/No |
| iv. | Is h_2 consistent? | Yes/No |
| v. | Is h_1 dominant h_2 ? | Yes/No |
| vi. | Is h_2 dominant h_1 ? | Yes/No |

Question No 07:

Consider the following map (not drawn to scale)

**Part a.**

- Use A* algorithm to work out a route from town A to town M. Use the following cost functions
- $g(n)$ = Total cost of reaching from town A to town n (step cost of each move is given on the map)
- $h(n)$ = Straight line distance between town n and town M These costs are given in the table below.
 $h(n)$ can be used as an estimated distance to M.

Town	Distance
A	56
B	22
C	30
D	29
E	29
F	30
G	14

Town	Distance
H	10
I	8
J	5
K	30
L	15
M	0

- Show the order of nodes expanded by A* with and without expanded list.
- What path/route would be found by A* using this heuristic function?